Xidong Wu

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Education

Northeastern University & University of Dundee
Undergraduate, Biomedical Engineering 2022 - 2026
GPA:88/100, First Class Honours

Main Publications

- **Wu, X.,** Yan, M., Wang, R., & Xie, L. Multiscale feature enhanced gating network for atrial fibrillation detection. *Computer Methods and Programs in Biomedicine (Q1 IF 6.4)*
- **Wu, X.,** Yan, M., Tang, H., Wu, D., & Xie, L. MSCGN: Multiscale complementary gating network for time series classification**.** *Biomedical Signal Processing and Control (Q1 IF 5.7)*
- **Wu, X.,** Chen H., Li Z., & Li, C. OG-SAM: Enhancing Multi-Organ Segmentation with Organogenesis-Based Adaptive Modeling. *Medical Image Computing and Computer Assisted Intervention Workshop (Oral)*
- Miao, Z.,**Wu, X.,** & Xie, L. MDF: A model dispatching framework with Noise-Estimator-Centric experts for robust biosignal classification. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*
- Tang, H., Yan M., **Wu, X.,** Lai, H., Zhang Y., Cui H., & Xie, L. Element-wise Product Enhanced Lightweight Model for Efficient ECG Arrhythmia Detection on Embedded Devices. *Computer Methods and Programs in Biomedicine (Q1 IF 6.4)*
- Tang, H., Cui, H., **Wu, X.,** & Xie, L. PhysioSAug: A novel data augmentation method based on diffusion probabilistic model for disease classification using physiological signals. *Expert Systems with Applications (Q1 IF 9.4)*
- **Wu, X.,** Tang, H., Wu, D., Jiang B., & Xie, L. Adaptive Aliasing Denoising for Noise-Robust Medical Time-Series Diagnosis**.** *Information Sciences (Q1 IF 6.0) Under Review*
- Wu, D., **Wu, X.,** Chen, M., Yu, C., & Xie, L. Time-varying and time-invariant feature bidirectional fusion network for recognizing unknown type of arrhythmia. *Neural Networks (Q1 IF 7.2) Under Review*
- Yan, M., Wang, H., . . . , **Wu, X.,** . . . , & Hong, S. ECG Memory Bank for Governable Use of Historical Cases in Diagnosis and Interpretation. *Submitted to Nature Biomedical Engineering.*

Research Experience

Multi Omics Driven Multi Organ Segmentation and Genomic Imaging Research
University of Cambridge, Faculty of Mathematics 04/2024 - Present

- Proposed OG-SAM (Organogenesis SAM), a multi organ segmentation framework based on organogenesis. By mimicking the organ differentiation pathway during embryonic development and dynamically adjusting the shared and unique feature representations between organs through parameter sharing and specialization, the accuracy and robustness of segmentation are improved.
- Design OG-SAM as a query based plugin that seamlessly integrates with SAM, using organ categories as queries to dynamically adjust the parameters of each organ, achieving precise segmentation for specific organs, especially when dealing with complex organ boundaries.

Heart-Anchored Electrophysiological Foundation Model for Precision Medicine
Northeastern University & University of Dundee, Faculty of Biomedical Engineering 03/2023 – Present

- Leveraged medical priors and multiscale gated architectures to enhance ECG-based atrial fibrillation detection and physiological time-series classification.
- Utilized diffusion models, adaptive denoising modules, and lightweight deployment strategies to address data imbalance, noise interference, and low-resource wearable applications.
- Designed a heart-anchored electrophysiological foundation model with dual-timescale representation learning for cross-dataset, cross-task, and cross-modality generalization toward precision medicine.

Technical Skills

TensorFlow, Keras, PyTorch, Python, Overleaf, Word